



April 25, 2019

10 CFR 50.73

SVP-19-024

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-29
NRC Docket No. 50-254

Subject: Licensee Event Report 254/2017-002-01, "Four Main Steam Isolation Valves (MSIVs) Closure Times Exceeded"

Enclosed is Licensee Event Report (LER) 254/2017-002-01, "Four Main Steam Isolation Valves (MSIVs) Closure Times Exceeded," for Quad Cities Nuclear Power Station, Unit 1.

This report is submitted in accordance with 10 CFR 50.73 (a)(2)(i)(B) which requires the reporting of any operation or condition prohibited by Technical Specifications.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Mr. Mark Humphrey at (309) 227-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "K. Ohr", written over a horizontal line.

Kenneth S. Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Quad Cities Nuclear Power Station Unit 1	2. Docket Number 05000254	3. Page 1 OF 4
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4. Title Four Main Steam Isolation Valves (MSIVs) Closure Times Exceeded
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	27	2017	2017	- 002	- 01	04	25	2019	N/A	N/A

9. Operating Mode 4	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 000	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER	
Licensee Contact Rachel Luebbe-Regulatory Assurance	Telephone Number (Include Area Code) (309) 227-2813

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
D	SB	HCU	A613	Y					

14. Supplemental Report Expected ☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No	15. Expected Submission Date Month: N/A Day: N/A Year: N/A
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Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On March 27, 2017, during refueling outage Q1R24 at 0840 hours, the Unit 1 Main Steam Isolation Valve (MSIV) as-found closure time test results indicated that four MSIVs failed to close within the required Technical Specification (TS) time of less than or equal to five seconds. The safety significance of this event was minimal.

The cause of two of the four MSIV slow closure timing events was inadequate procedure on how to apply Super O-Lube to valve stem bottom O-ring of the airpack manifold solenoid, and the cause of the other two slow closure timing events was environmental conditions in the Unit 1 MSIV Room.

Corrective actions included replacing the 1-0203-1B and 1-0203-1D MSIV airpack manifolds in Q1R25, revising the MSIV airpack manifold maintenance procedure regarding the amount of Super O-Lube to be applied, and revising operating procedures to prevent MSIV closure time drift observed to occur at MSIV room temperatures below 90 degrees F.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), which requires reporting of any operation or condition that was prohibited by the plant's TS.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities Station Unit 1	05000254	2017	- 002	- 01

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Four Main Steam Isolation Valves (MSIVs) Closure Times Exceed Technical Specifications Limit

A. CONDITION PRIOR TO EVENT

Unit: 1 Event Date: March 27, 2017 Event Time: 0840 hours
Reactor Mode: 4 Mode Name: Cold Shutdown Power Level: 0%

B. DESCRIPTION OF EVENT

On March 27, 2017, while Unit 1 was in Mode 4 for refueling outage Q1R24, Operations was performing surveillance, "MSIV Closure Timing," when four of the eight Unit 1 Main Steam [SB] Isolation Valves [ISV] (MSIVs), 1-0203-1B, 1-0203-1D, 1-0203-2C, and 1-0203-2D failed to close within the required cold shutdown Technical Specification (TS) limit of greater than or equal to three seconds and less than or equal to five seconds. The closure times for those four MSIVs were 5.7, 5.2, 5.1, and 5.6 seconds, respectively. The other four MSIVs all closed within the required TS time.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), which requires reporting of any operation or condition that was prohibited by the plant's Technical Specifications.

C. CAUSE OF EVENT

MSIVs 1-0203-2C and 1-0203-2D are located in the Unit 1 MSIV Room and MSIVs 1-0203-1B and 1-0203-1D are located in the Unit 1 Drywell. The cause of the slow closure timing events for the 1-0203-1B and 1-0203-1D MSIVs was due to an inadequate procedure on how to apply Super O-Lube to valve stem bottom O-ring of the airpack manifold. The slow closure times were due to excess Super O-Lube on airpack manifold solenoids. The maintenance procedure for applying the Super O-Lube was not specific on the amount of lubricant to be used on the bottom O-ring.

The cause of the slow closure timing events for the 1-0203-2C and 1-0203-2D MSIVs was due to environmental conditions in the Unit 1 MSIV Room. Beginning in June 2009, the average temperature in the Unit 1 MSIV room was significantly reduced due to the replacement of the MSIV room coolers. An abnormal drift between the as-left closure time from the previous outage to the as-found closure time of the current outage has been observed to be correlated with cooler room temperatures. A similar temperature-correlated drift was not observed for the MSIVs located inside the drywell, where the temperature is consistently warmer than the MSIV Room.

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2. DOCKET NUMBER

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2017	- 002	- 01

NARRATIVE**D. SAFETY ANALYSIS****System Design**

The design of the MSIVs is to prevent reactor coolant [AD] inventory loss and protect plant personnel in the event of steam line breakage outside the isolation valves, and to complete the primary containment [NH] boundary after a Loss of Coolant Accident (LOCA). The MSIVs are 20-inch air/spring operated, balanced "Y"-type globe valves. There are four main steam lines and each steam line has two isolation valves, one inside (inboard MSIV) and one outside (outboard MSIV) of primary containment.

Updated Final Safety Analysis Report (UFSAR) Section 6.2.4.1 provides the valve closure time for the main steam line is based on the main steam line break accident discussed in Section 15.6. Ensuring the MSIV closure time less than or equal to five seconds, sufficient coolant will remain in the reactor vessel to provide adequate core cooling. The valves are designed to close and to be leak-tight during the worst conditions of pressure, temperature, and steam flow following a break in the main steam line outside the containment. The MSIVs are leak tested in accordance with the 10 CFR 50 Appendix J program. UFSAR Section 15.6 provides that a maximum MSIV closure time of 10.5 seconds would limit the total amount of liquid and steam lost from the primary system to prevent the core from being uncovered, and this closure time will ensure the radiological doses are well below the guidelines set forth in 10 CFR 100.

Proper adjustment of the MSIV closure times prevents operation outside of the operational and design limits.

Safety Impact

Technical Specifications and the In-Service Testing program require the MSIVs to close in a three to five second time range. Five seconds is fast enough to prevent a gross release of fission products, and three seconds is slow enough to minimize the severity of the pressure transient resulting from isolating the main steam lines during full power operation. Slow MSIV stroke times would challenge the steam release limits of a steam line break outside of containment should the corresponding inboard valve simultaneously fail to close.

The Unit 1 MSIVs 1-0203-1B, 1-0203-1D, 1-0203-2C, and 1-0203-2D failed to close within the required Technical Specification limit of less than or equal to five seconds. The closure times for those four MSIVs were 5.7, 5.2, 5.1, and 5.6 seconds, respectively, however, the four MSIVs all closed inside the UFSAR limit of less than or equal to 10.5 seconds. Although the required five second Technical Specification limit was not met for these four MSIVs, the MSIVs were capable of performing the required UFSAR safety function during the operating cycle prior to Q1R24. Furthermore, since the MSIVs were not required to be operable or available at the time of discovery, this condition did not create any actual plant or safety consequences since the unit was not in an accident or transient condition requiring the use of main steam line isolation valves during this period of time.

Risk Insights

The closure times for the four MSIVs were 5.7, 5.2, 5.1, and 5.6 seconds and recorded as slow, but the MSIVs did fully close. The MSIV success criterion in the Plant Probabilistic Risk Assessment (PRA) model, however, does not depend on closure time. Therefore, a difference of a fraction of a second between actual MSIV closure time and the acceptance criterion for the Technical Specification closure time has no effect on the PRA model. As a result, there is no effect on Core Damage Frequency (CDF) or Large Early Release Frequency (LERF).

In conclusion, the failure of the four MSIVs to close within the required time would not have caused the station to exceed the inventory and dose release limits. Furthermore, the impact on risk of this event is negligible. Therefore, the overall safety significance of this event was minimal.

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NARRATIVE**E. CORRECTIVE ACTIONS**

Immediate:

1. The 1-0203-1B and 1-0203-1D MSIV airpack manifolds were replaced during the outage, Q1R25 in 2019.
2. Procedures concerning the MSIV Room Cooler operation have been revised to maintain a consist temperature in the MSIV Room.

Follow-up:

1. The MSIV lubrication surveillance will be revised to provide additional information on the amount of Super O-Lube to be used in the airpack manifold solenoid O-rings.

F. PREVIOUS OCCURRENCES

The station events database, LERs, and INPO ICES were reviewed for similar events at Quad Cities. Specifically, this event was primarily attributed to an inadequate procedure regarding the amount of Super O-Lube on the airpack manifold bottom O-rings and cooler temperatures in the MSIV Room. Based on the causes of this event and associated corrective actions, the events listed below, provided insight into resolution of this event.

- Station Issue Report (IR) 1213432, Unit 1 Outboard MSIV 1-0203-2C Failed QCOS 0250-04 (05/09/2011) - The 1-0203-2C outboard MSIV closure time exceeded the acceptance criteria for cold timing, which is less than or equal to five seconds. The stroke time recorded was 5.2 seconds. The cause was dirty contacts within the control circuitry creating an electrical time delay. This event occurred after the MSIV Room environmental conditions changed, which led to the current action to control the temperature in the MSIV Room. Therefore, IR 1213432 provided useful insights associated with this event.
- LER 2013-002-00 Outboard Main Steam Isolation Valves (MSIVs) Stroke Times Exceeded (03/11/2013) - All four Unit 1 outboard MSIVs closed slowly due to age and wear degraded actuator seals. The vendor recommended a replacement frequency of eight years for the actuators. This event was the second occurrence of slow closure times after the MSIV Room environmental conditions changed, which led to the current action to control the temperature in the MSIV Room. Therefore, LER 2013-002-00 provided useful insights associated with this event.

G. COMPONENT FAILURE DATA

The airpack manifold solenoid is manufactured by Automatic Valve Nuclear (formerly AVCO) [A613] model # C6930-010.

This event has been reported to ICES.